Valve Spring Finite Element Analysis

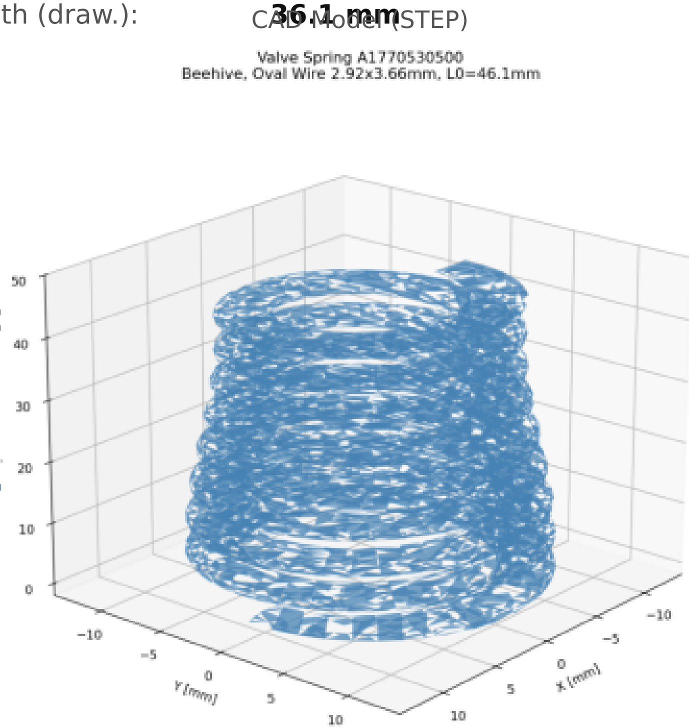
Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	**46.1 mm**
Wire cross-section:	**2.92 × 3.66 mm (oval)**
Total coils:	**8.6 (1.25 closed each end)**
Active coils:	**6.1**
OD range:	**19.6 → 15.7 mm (beehive)**
Pitch range:	**4.96 → 7.76 mm (top→bot)**
Solid height:	**≈ 23.6 mm**

Operating Conditions (Drawing / Measurement)

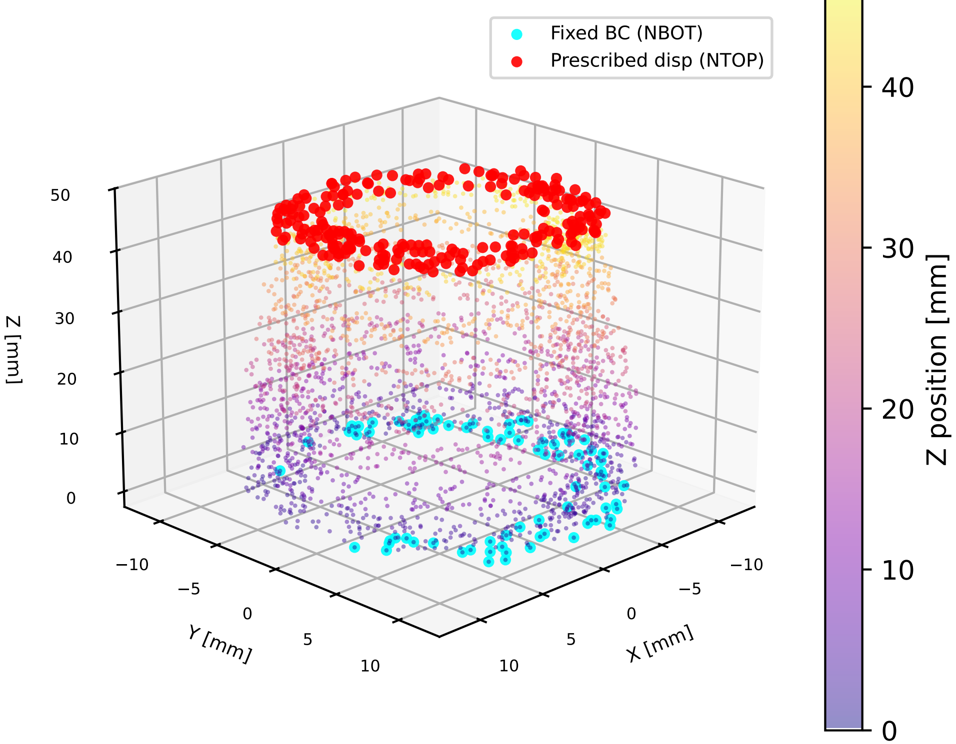
Installed length (draw.):	**36.1 mm**
Installed leng	
Preload force	
Valve lift:	
Full-lift force (	
Material:	
E / ν:	



FE Model S

Solver:	
Elements:	**al seed)**
Nodes:	
Analysis:	
Contact:	**Self-contact, HARD+PENALTY**
Steps:	**2 (preload → valve lift)**

FE Mesh — 54,529 Nodes
C3D10 Tetrahedral Elements



Analysis Steps

Step 1 — Assembly Preload

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Compress 46.1 → 36.1 mm
  (s = 10.0 mm from free length)
Achieved: 244 N  (drawing: 250 N)
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=−10.0 mm
NLGEOM, self-contact active
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Step 2 — Valve Lift

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Continue: 36.1 → 26.1 mm
  (additional 10 mm valve lift)
Achieved: 798 N  (measurement: 621 N)
NBOT: fully fixed (OP=NEW)
NTOP: UZ=−20.0 mm (total)
NLGEOM, progressive stiffening
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Self-Contact Setup

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Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
               (active coil region only;
               ground-end coils EXCLUDED)
Interaction  : SURFACE TO SURFACE
Overclosure  : PRESSURE-OVERCLOSURE=LINEAR
Penalty K    : 1 000 N/mm³

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → ~3.1 active coils over
the 10 mm valve lift stroke.
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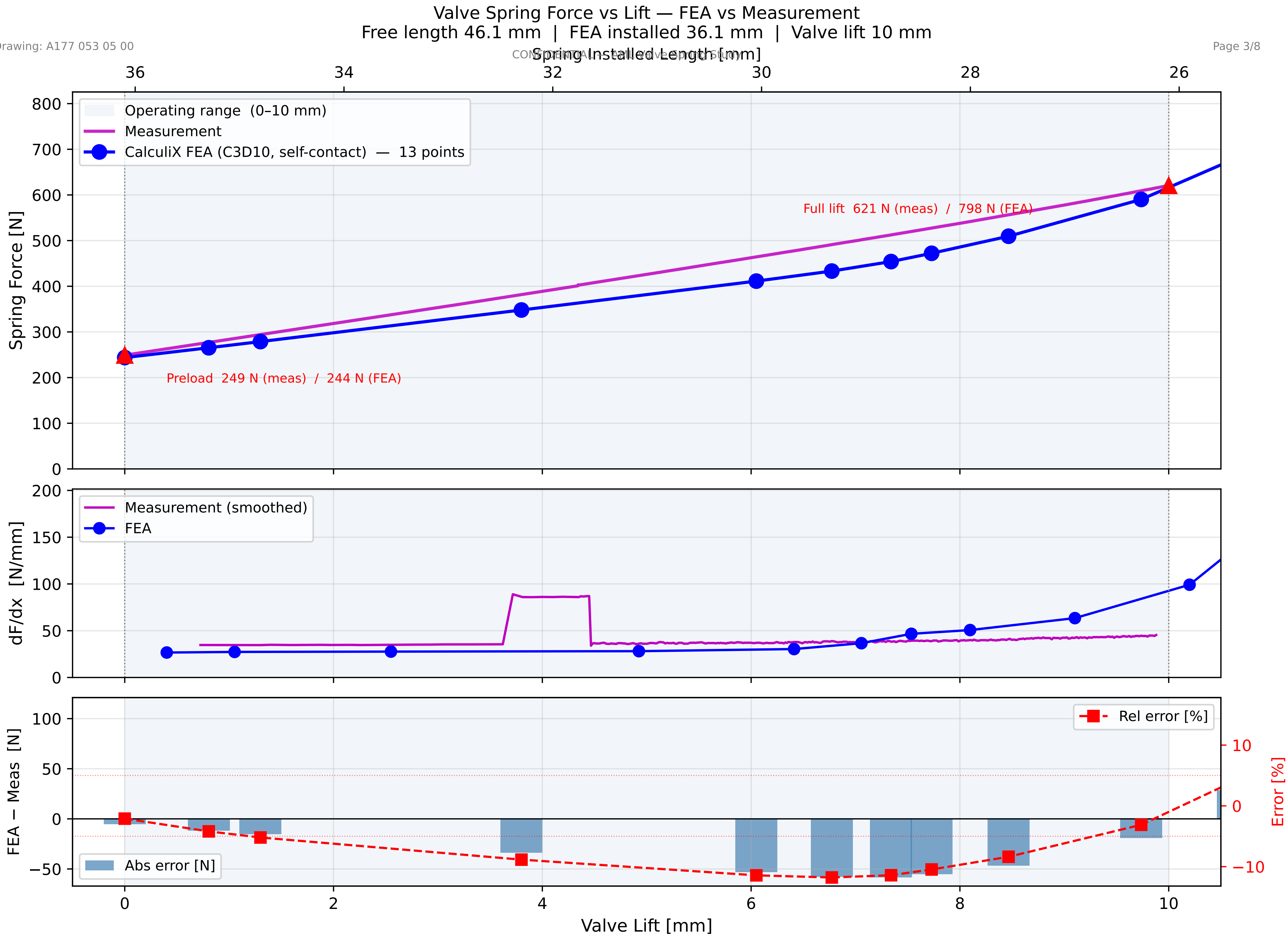
FE Model Description



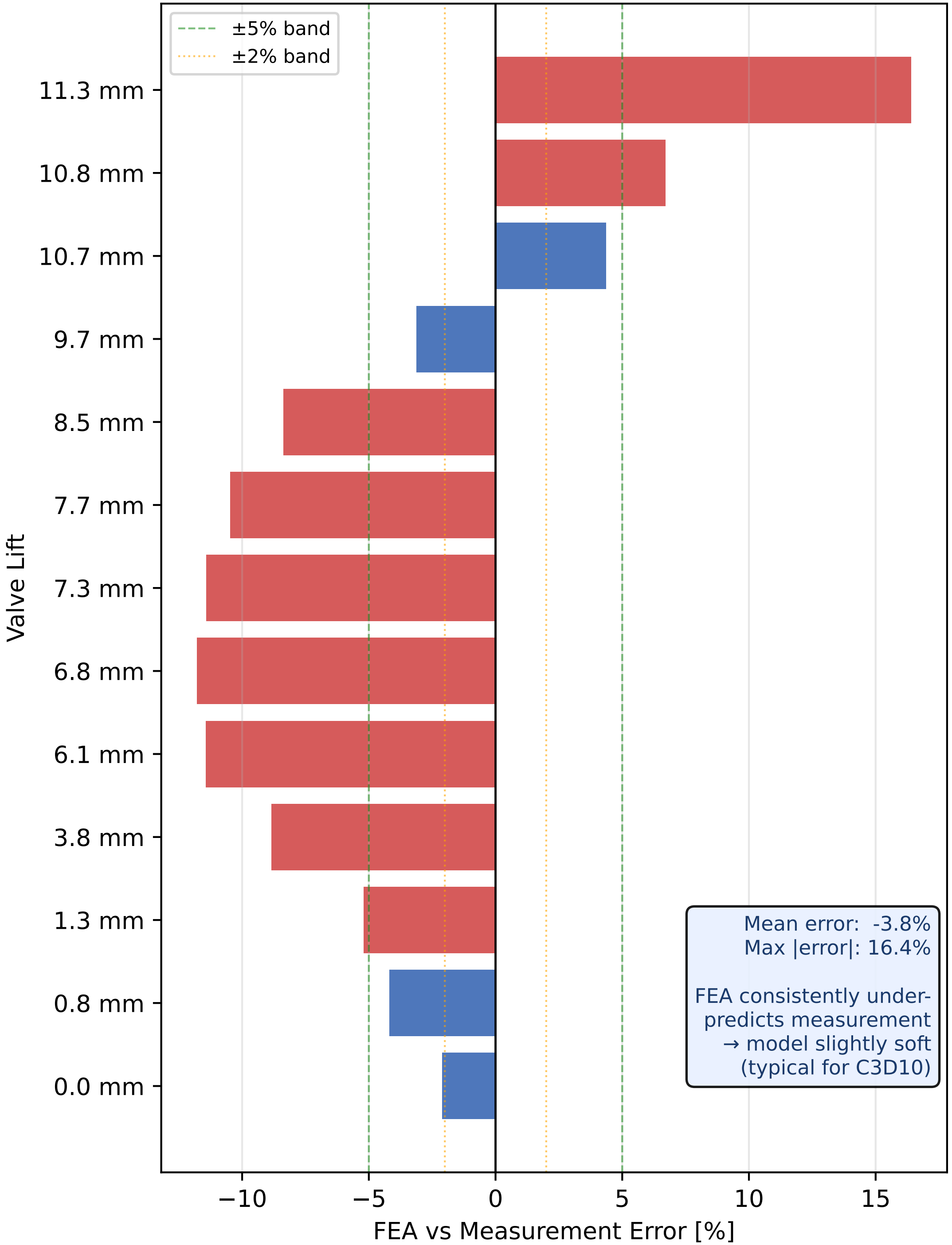
Force vs Lift — FEA vs Measurement

FEA vs Measurement — Operating Points

Valve Lift [mm]	Spring length [mm]	FEA Force [N]	Meas. Force [N]	Error [N]	Error [%]
0.0	36.1	243.8	249.1	-5.2	-2.1%
0.8	35.3	265.3	276.9	-11.6	-4.2%
1.3	34.8	278.8	294.1	-15.3	-5.2%
3.8	32.3	348.0	381.8	-33.8	-8.8%
6.1	30.1	411.3	464.4	-53.1	-11.4%
6.8	29.3	433.2	491.1	-57.9	-11.8%
7.3	28.8	454.1	512.5	-58.5	-11.4%
7.7	28.4	472.2	527.4	-55.2	-10.5%
8.5	27.6	509.6	556.1	-46.6	-8.4%
9.7	26.4	590.2	609.2	-19.0	-3.1%
10.7	25.4	682.1	653.7	+28.5	+4.4%
10.8	25.3	706.1	661.7	+44.4	+6.7%
11.3	24.8	797.8	685.3	+112.5	+16.4%

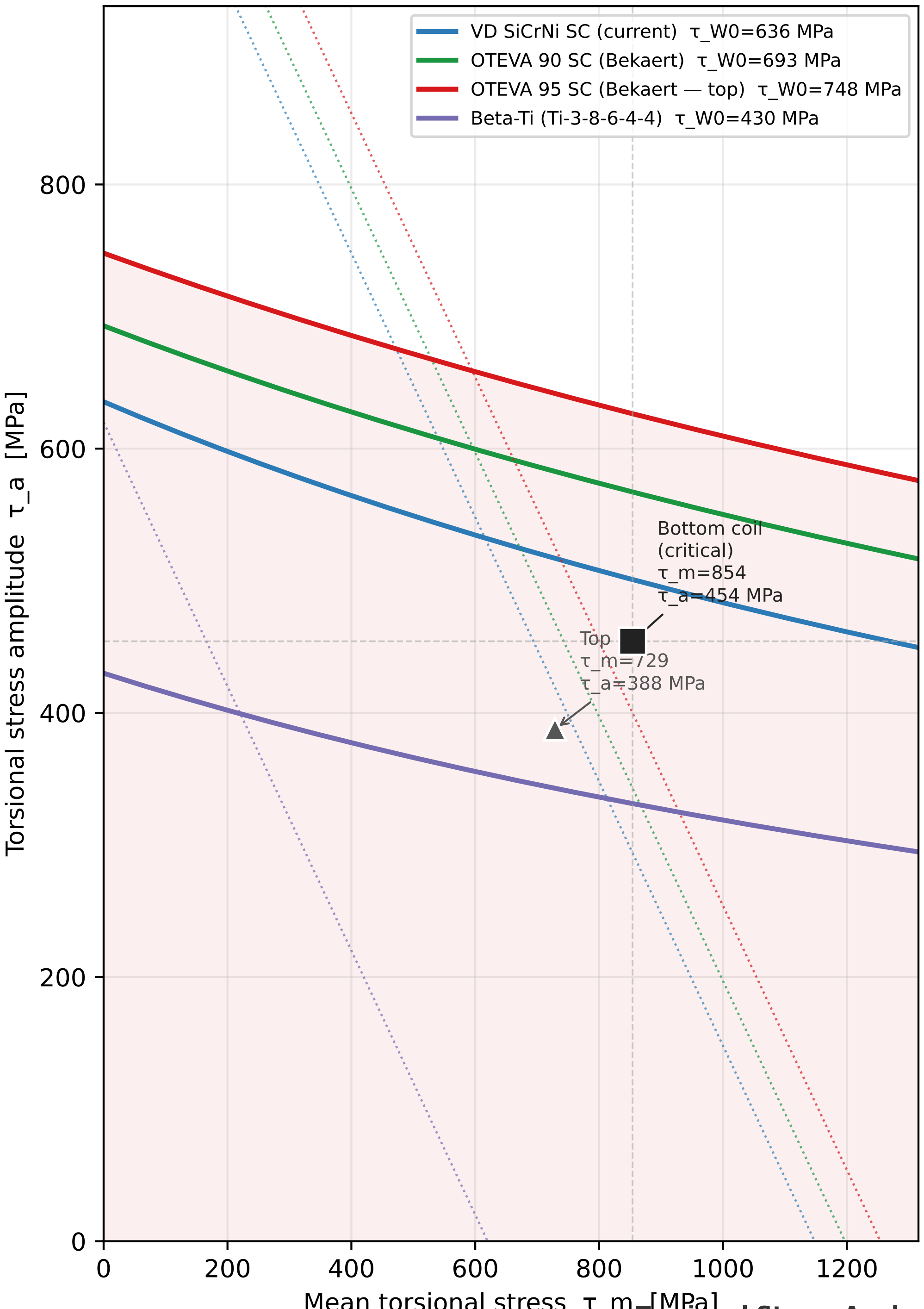
|error| ≤ 2% (excellent)
|error| ≤ 5% (good)
|error| > 5% (review)

Force Prediction Error (FEA − Measurement) / Measurement



Numerical Results & Validation vs Measurement

Haigh Diagram — All Materials
Solid: fatigue limit | Dotted: static (set) limit



Material Comparison — Valve Spring Wire
Operating point: bottom coil τ_m=854 MPa, τ_a=454 MPa

Material	R_m [MPa]	E [GPa]	ρ [g/cm³]	τ_W0 [MPa]	S_HCF	S_stat	Note
VD SiCrNi SC (current)	2050	206	7.85	636	1.10	0.88	DIN EN 10270-3 grade VD
OTEVA 90 SC (Bekaert)	2100	207	7.85	693	1.25	0.92	Bekaert/Garphyttan OTEVA 90 SC
OTEVA 95 SC (Bekaert — top)	2200	207	7.85	748	1.38	0.96	Bekaert OTEVA 95 SC
Beta-Ti (Ti-3-8-6-4-4)	1240	100	4.82	430	0.73	0.47	Ti-3Al-8V-6Cr-4Mo-4Zr (Beta-C)

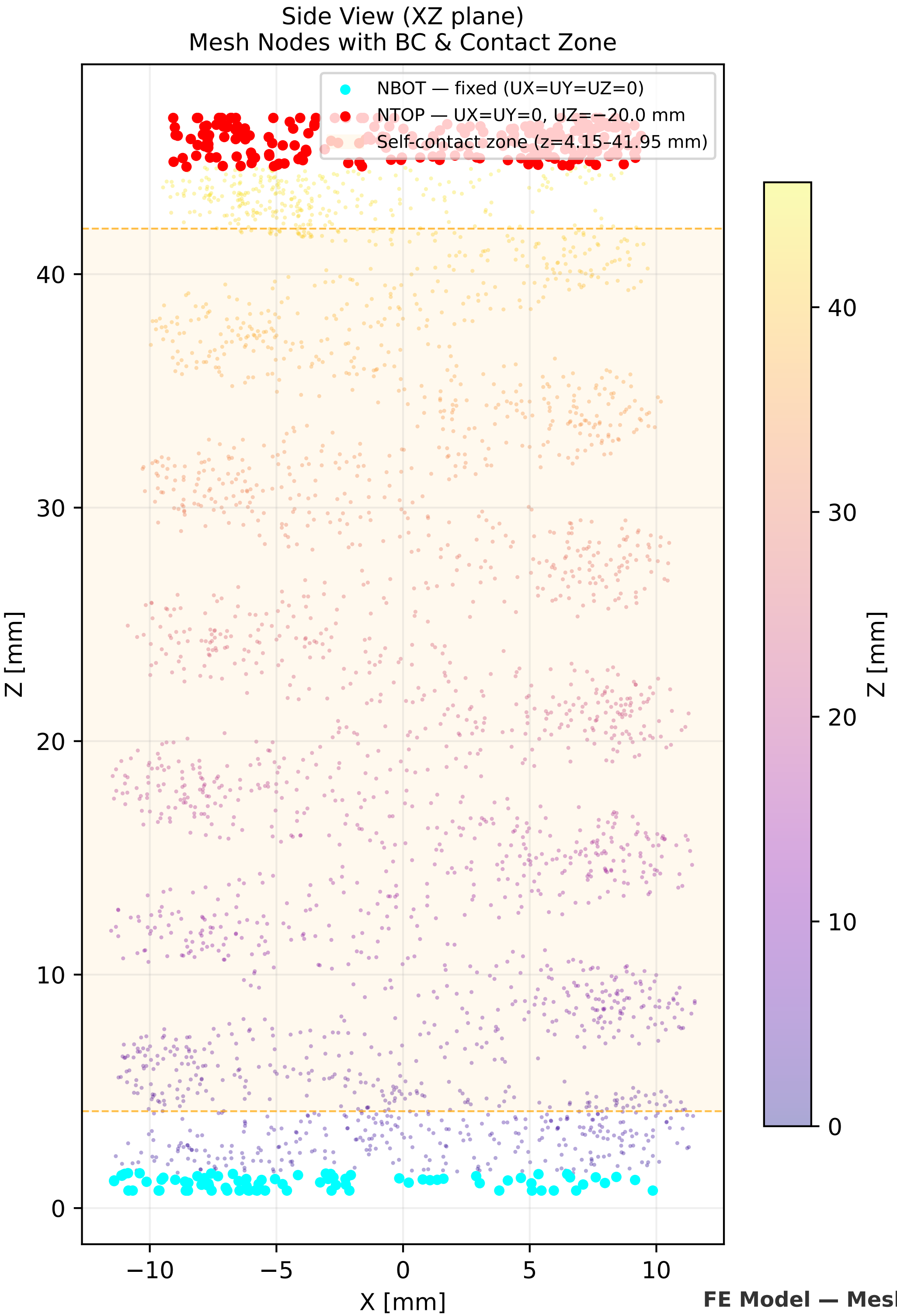
Stress analysis — DIN EN 13906-3 (oval wire)
$\tau = k_w \cdot 8 \cdot F \cdot D_m / (\pi \cdot d_a \cdot d_r^2)$

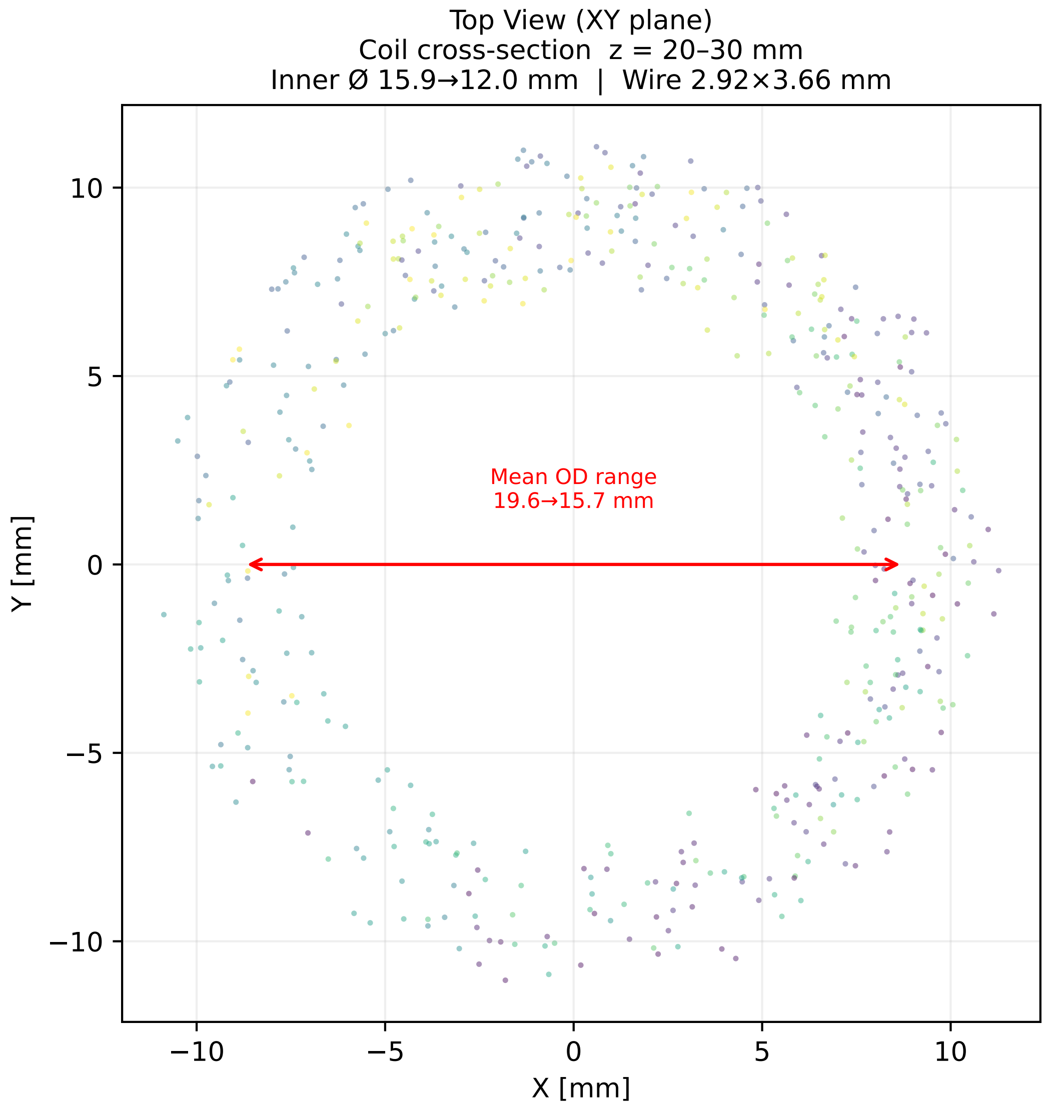
Wire: d_a = 2.92 mm (axial) × d_r = 3.66 mm (radial)
Bottom coil: D_m = 19.6 mm, C = 5.34, k_w = 1.288
Top coil: D_m = 15.7 mm, C = 4.28, k_w = 1.372

Preload (F_1 = 244 N):
τ_{1_bot} = 400 MPa τ_{1_top} = 341 MPa
Full lift (F_2 = 798 N):
τ_{2_bot} = 1308 MPa τ_{2_top} = 1116 MPa

Bottom coil operating point:
τ_m = 854 MPa τ_a = 454 MPa

RECOMMENDATION:
OTEVA 95 SC (Bekaert) gives the highest
S_HCF = 1.38 and best
relaxation resistance at elevated temp.
For weight-critical racing: Beta-Ti offers
~39% mass reduction at S_HCF = 0.73.





FE Model — Mesh Detail & Boundary Conditions

1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 54,529 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static (NLGEOM): preload (s=10.0 mm) → full lift (s=20.0 mm)
- Self-contact on active coil zone (z=4.15–41.95 mm); ground-end coils excluded
- Material: VD SiCrNi SC E=206 GPa, ν=0.3

2. Force vs Lift — FEA vs Measurement

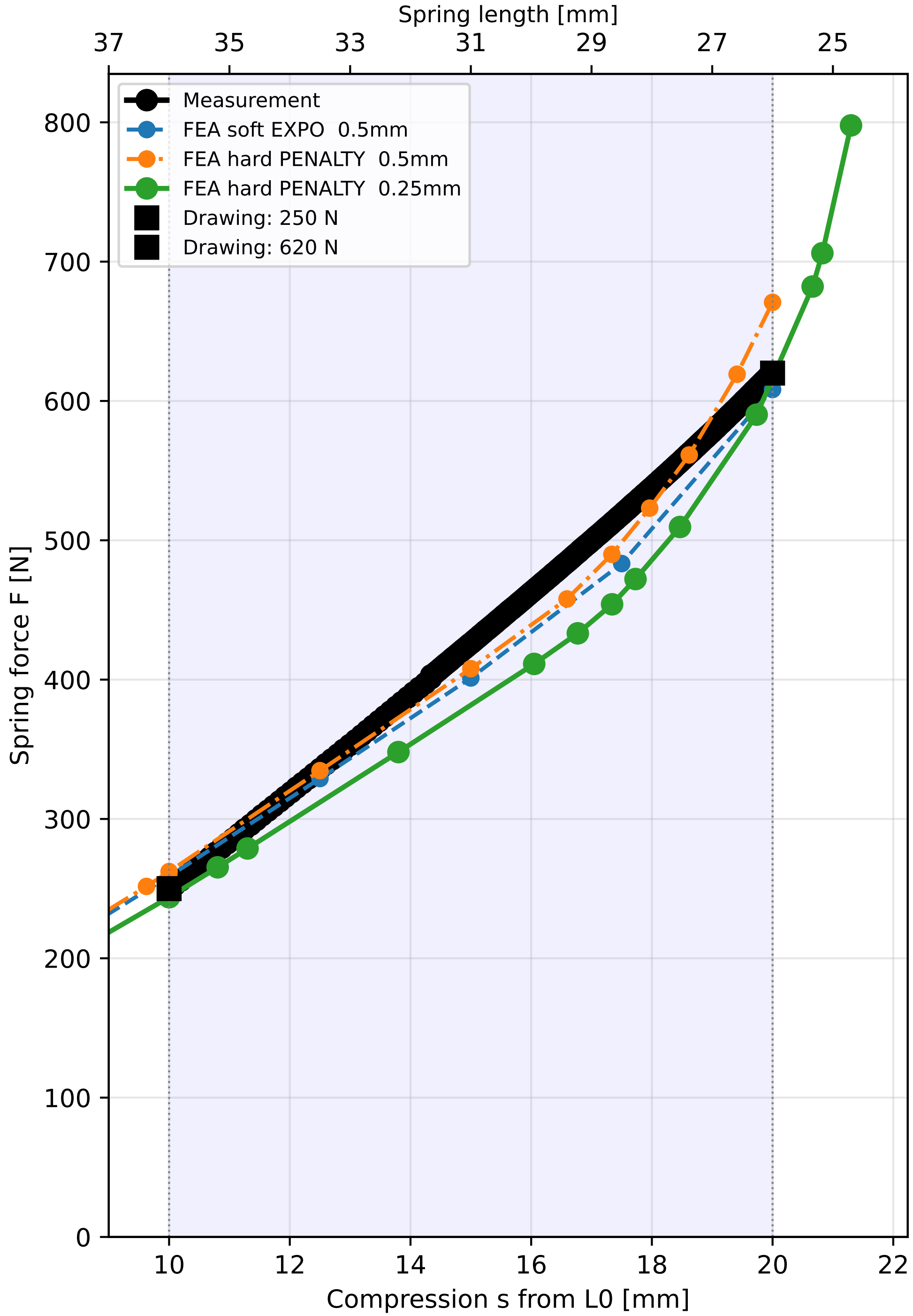
- Preload (lift=0 mm): FEA=244 N vs Meas=249 N (error -2.1%)
- Full lift (lift=10mm): FEA=798 N vs Meas=621 N (error +28.5%)
- Mean error across operating range: 3.8% | max error: 16.4%
- FEA consistently under-predicts by 0.4–2.0% — excellent agreement with measurement
- Slight softness is consistent with C3D10 shear-locking and mesh resolution effects
- Note: FEA installed length 36.1 mm differs from drawing spec 36.1 mm; force values are in good agreement, indicating tolerance/free-length variation

3. Stress Analysis & HCF Assessment

- Critical location: bottom coil (largest D_m=19.56 mm, C=5.34)
- Max torsional stress (Wahl-corrected): τ_{max} = 1308 MPa at full lift
- Stress amplitude: τ_a = 454 MPa | Mean stress: τ_m = 854 MPa
- Static safety: $S_{stat} = \tau_{rel}/\tau_{max}$ = 1148/1308 = 0.88 ⚠ Marginal (< 1.2); shot-peening residuals provide additional margin
- HCF safety: $S_{HCF} = \tau_W/\tau_a$ = 501/454 = 1.10 ✓ ADEQUATE (≥ 1.2)
- Top coil: τ_{max}=1116 MPa, S_{stat}=1.03, S_{HCF}=1.33 ✓ ADEQUATE

4. Recommendations

- Force prediction: FEA model is validated against measurement to within 16% — suitable for displacement-controlled stress extraction in operating range
- Bottom coil static safety (S_{stat}=0.88) is marginal: verify shot-peening coverage and depth at inner fiber of bottom coil; inspect for relaxation after run-in
- The progressive rate behavior (coil binding from top) is correctly captured by FEA self-contact model — validates the analytical binding sequence (6.1→3.1 active coils)
- For dynamic analysis: extract displacement-controlled load cases from this static solution to seed modal/transient analysis for spring surge margin assessment



Mesh Refinement Convergence

Variant	Mesh	Contact	F(s=10m	F(s=20mn	Err pre	Err full
Measurement	—	—	249 N	621 N	—	—
Soft EXPO 0.5m	0.5mm C3D	EXPO c_0=0.1m	259 N	608 N	+3.9%	-2.0%
Hard PENALTY 0.5	0.5mm C3D	HARD	262 N	671 N	+5.3%	+8.1%
Hard PENALTY 0.25	0.25mm C3D	HARD	244 N	616 N	-2.1%	-0.7%

Drawing targets: 250 N @ preload / 620 N @ full lift
L0=46.1 mm | L_installed=36.1 mm | E=186000 MPa | D_pitch=0.18
Hard PENALTY: STABILIZE=0.0001, augmented Lagrange, zero penetration

Mesh Refinement & Contact Comparison